

DESIGN AND ANALYSIS OF REGENERATIVE BRAKING SYSTEM IN ELECTRIC AND HYBRID ELECTRIC VEHICLES

A Major Project Report

Submitted by

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Under the guidance of

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This is to certify that the Major Project work entitled “**Design and Analysis of Regenerative Braking System in Electric and Hybrid Electric Vehicles**” is carried out by **Abhishek Kumar Sadhan (22110117013)** and in partial fulfillment for the award of degree of **Bachelor of Technology in Electrical & Electronics Engineering**, LoknayaK Jai Prakash Institute of Technology Chapra, Affiliated to Bihar Engineering University, Patna during the academic year 2022–2026.

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Declaration

I hereby declare that the project titled “**DESIGN AND ANALYSIS OF REGENERATIVE BRAKING SYSTEM IN ELECTRIC AND HYBRID ELECTRIC VEHICLES**” submitted to Loknayak Jai Prakash Institute of Technology Chapra, affiliated to Bihar Engineering University (BEU), Patna for the award of the degree of Bachelor of Technology in Electrical & Electronics Engineering is a result of original research carried out in this report. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of degree or diploma.

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Abstract

The growing demand for energy-efficient and environmentally friendly transportation has led to rapid advancements in electric vehicle (EV) and hybrid electric vehicle (HEV) technologies. One of the most critical challenges in these vehicles is the efficient management of energy, particularly the energy dissipated during braking, which in conventional friction-based systems is entirely lost as heat. Regenerative braking systems offer a practical and effective solution to this problem by converting the kinetic energy of a decelerating vehicle into electrical energy, which is then stored in the battery for reuse.

This project presents a comprehensive study of the design and analysis of regenerative braking systems in electric and hybrid electric vehicles. The working principle is based on the dual-mode operation of an electric motor, which acts as a motor during propulsion and as a generator during deceleration. The recovered electrical energy is conditioned using power electronic components including rectifiers, MOSFET-based control circuits, and DC-DC converters to ensure stable voltage regulation and safe battery charging.

The study covers the fundamental concepts of regenerative braking, the classification of hybrid electric vehicle architectures, and various braking strategies such as parallel and series regenerative braking. Key factors influencing energy recovery efficiency — including vehicle speed, battery state of charge (SOC), motor characteristics, and driving conditions — have also been examined. A detailed mathematical modelling of the energy recovery process, thermal analysis of braking components, and a techno-economic feasibility assessment are presented as extended contributions of this work. The simulation results demonstrate that the proposed system achieves a significant reduction in energy wastage and improves the overall driving range of the vehicle by approximately 17%. The system also reduces mechanical wear on friction braking components, leading to lower maintenance costs and improved component lifespan.

Keywords: Regenerative Braking, Electric Vehicles, Hybrid Electric Vehicles, Energy Recovery, DC Motor, Power Electronics, Battery Charging, SOC, MOSFET, Dual-Mode Operation, Energy Efficiency, Thermal Analysis, Mathematical Modelling.

Contents

1	Introduction	1
1.1	Introduction	1
1.2	Problem Statement	2
1.3	Objectives	2
1.4	Scope of the Work	3
1.5	Thesis Overview	3
2	Literature Survey	5
2.1	Introduction	5
2.1.1	Regenerative Braking on Electric Vehicles: Working Principles and Benefits	5
2.1.2	Hybrid Electric Vehicles and Regenerative Braking	5
2.1.3	Energy Efficient Electric Vehicle Using Regenerative Braking System	6
2.1.4	Design and Analysis of Regenerative Braking System in Electric Vehicles	6
2.1.5	Research on Regenerative Braking Strategies for Electric Vehicles	6
2.1.6	Integration of Regenerative Braking with Vehicle Safety Systems	6
2.1.7	Energy Recovery Efficiency: Influence of Battery Management	7
2.1.8	Advanced Motor Technologies for Regenerative Braking	7
2.1.9	Economic and Environmental Impact	7
2.1.10	Summary	8
3	Electric and Hybrid Electric Vehicles	9
3.1	Introduction to Electric Vehicles	9
3.2	Classification of Hybrid Electric Vehicles	9
3.2.1	Series Hybrid Electric Vehicle (SHEV)	10
3.2.2	Parallel Hybrid Electric Vehicle (PHEV)	10
3.2.3	Series-Parallel Hybrid Electric Vehicle (SP-HEV)	10
3.3	Comparison of HEV Architectures	11
3.4	Characteristics of HEVs	11
3.5	Current Development Status of HEVs	12
3.5.1	Technology Development	12
3.5.2	Existing Challenges	12
3.6	Types of Braking Systems	12
3.6.1	Frictional Braking	12

3.6.2	Electromagnetic Braking	13
3.6.3	Regenerative Braking	13
4	Regenerative Braking System	14
4.1	Fundamentals of Regenerative Braking	14
4.2	Law of Conservation of Energy	14
4.3	Components of the Regenerative Braking System	15
4.4	Principle of Operation	16
4.4.1	Driving Mode	16
4.4.2	Braking Mode	16
4.5	Categories of Regenerative Braking	16
4.5.1	Category A – Compression Regenerative Braking	16
4.5.2	Category B – Service Braking Regenerative Braking	17
4.6	Regenerative Braking Strategies	17
4.6.1	Parallel Strategy	17
4.6.2	Series Strategy	17
4.7	Energy Losses in Regenerative Braking	17
4.8	Factors Influencing Energy Recovery	18
5	Design and Methodology	19
5.1	System Architecture	19
5.2	Components Used	19
5.3	Energy Flow Design	20
5.3.1	Driving Mode Operation	20
5.3.2	Braking Mode Operation	21
5.4	Control Mechanism	21
5.5	Power Conditioning and Storage	23
5.6	Energy Storage Systems	23
5.6.1	Batteries	23
5.6.2	Ultracapacitors / Supercapacitors	23
6	Mathematical Modelling of the Regenerative Braking System	25
6.1	Introduction	25
6.2	Vehicle Dynamics Model	25
6.2.1	Kinetic Energy Available for Recovery	25
6.2.2	Braking Force Distribution	25
6.2.3	Deceleration Profile	26
6.3	Motor-Generator Electromechanical Model	26
6.3.1	DC Motor in Generator Mode	26
6.3.2	Generated Power	27
6.3.3	Generator Torque	27

6.4	Power Electronics Model	27
6.4.1	Bridge Rectifier	27
6.4.2	Buck (Step-Down) DC-DC Converter	28
6.5	Battery Charging Model	28
6.5.1	Equivalent Circuit Model	28
6.5.2	State of Charge Update	29
6.6	Overall System Energy Recovery Efficiency	29
6.7	Energy Recovery per Braking Event – Example Calculation	29
7	Simulation Results and Discussion	31
7.1	Simulation Setup	31
7.2	Results of Regenerative Braking System	31
7.2.1	Motor Mode Switching	31
7.2.2	Voltage Generation	31
7.2.3	Battery Recharging	32
7.2.4	System Efficiency	33
7.3	Comparison of Braking Strategies	33
7.4	Energy Recovery Analysis	34
7.5	SOC Evolution During Driving Cycle	34
7.6	Discussion	35
8	Thermal Analysis and Techno-Economic Feasibility	36
8.1	Introduction	36
8.2	Thermal Analysis of Braking Components	36
8.2.1	Background	36
8.2.2	Heat Generation in Friction Brakes	36
8.2.3	Brake Disc Temperature Rise	37
8.2.4	Motor Thermal Behaviour During Regeneration	37
8.2.5	Summary of Thermal Analysis	37
8.3	Techno-Economic Feasibility Analysis	38
8.3.1	Introduction	38
8.3.2	Operating Cost Savings	38
8.3.3	Brake Maintenance Savings	38
8.3.4	Total Annual Savings and Payback Period	39
8.3.5	Environmental Benefit Quantification	39
9	Conclusion and Future Scope	41
9.1	Conclusion	41
9.2	Future Scope	42
	Bibliography	45

List of Figures

3.1	Block Diagram of a Basic Electric Vehicle Drivetrain	9
3.2	Structure of Series Hybrid Electric Vehicle (SHEV)	10
3.3	Structure of Parallel Hybrid Electric Vehicle (PHEV)	10
4.1	Principle of Regenerative Braking – Energy Conversion Flow	15
5.1	System Architecture of Regenerative Braking System	19
5.2	Flowchart of the Regenerative Braking System Control Logic	22
7.1	Generated Voltage and Charging Current during Regenerative Braking Operation	32
7.2	Energy Recovery Rate Comparison for Different Braking Strategies	33
7.3	Battery SOC Evolution During Simulated 10 km Urban Drive Cycle	34

List of Tables

3.1	Comparison of HEV Architectures	11
4.1	Factors Influencing Regenerative Braking Energy Recovery	18
5.1	Components Used in Regenerative Braking System	20
5.2	SOC-based Regenerative Braking Control Logic	23
7.1	Generated Voltage and Charging Current vs Motor Speed	32
7.2	Energy Recovery Rate Comparison – Parallel vs Series Strategies	33
7.3	Energy Analysis – With and Without Regenerative Braking	34
8.1	Thermal Analysis Summary – Conventional vs Regenerative Braking	38
8.2	Techno-Economic Summary of Regenerative Braking Implementation	40
9.1	Comparison of Energy Recovery – Conventional vs Regenerative Braking	42

List of Abbreviations

Abbreviation	Full Form
ABS	Anti-lock Braking System
AC	Alternating Current
BEMF	Back Electromotive Force
BEV	Battery Electric Vehicle
BMS	Battery Management System
DC	Direct Current
EDLC	Electrochemical Double Layer Capacitor
EMF	Electromotive Force
ESC	Electronic Stability Control
EV	Electric Vehicle
GHG	Greenhouse Gas
HEV	Hybrid Electric Vehicle
ICE	Internal Combustion Engine
IoT	Internet of Things
LCD	Liquid Crystal Display
M/G	Motor/Generator
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
PCB	Printed Circuit Board
PHEV	Parallel Hybrid Electric Vehicle
PMSM	Permanent Magnet Synchronous Motor
PWM	Pulse Width Modulation
RBS	Regenerative Braking System
SHEV	Series Hybrid Electric Vehicle
SOC	State of Charge
SP-HEV	Series-Parallel Hybrid Electric Vehicle
SRM	Switched Reluctance Motor
BLDC	Brushless Direct Current Motor
FEA	Finite Element Analysis
NPC	Net Present Cost
IRR	Internal Rate of Return
NPV	Net Present Value
PID	Proportional-Integral-Derivative (Controller)
WLTP	Worldwide Harmonised Light Vehicle Test Procedure
UDDS	Urban Dynamometer Driving Schedule

Chapter 1: Introduction

1.1 Introduction

The rapid growth in global industrialization, urbanization, and transportation demands has brought with it a critical challenge: the dependence on fossil fuels and the resulting environmental pollution. Conventional vehicles powered by internal combustion engines (ICE) are major contributors to greenhouse gas (GHG) emissions, air pollution, and the depletion of non-renewable energy sources. In response to these challenges, the automotive industry has been actively pursuing cleaner, more efficient alternatives, with electric vehicles (EVs) and hybrid electric vehicles (HEVs) emerging as the most promising solutions [2].

Electric vehicles operate entirely on electrical energy stored in batteries, while hybrid electric vehicles combine an internal combustion engine with one or more electric motors. Both types of vehicles offer significant advantages over conventional vehicles in terms of emissions and fuel efficiency. However, the limited driving range of electric vehicles, primarily due to battery capacity constraints, remains one of the most significant challenges in the widespread adoption of this technology.

One of the most effective methods to address this challenge is the implementation of regenerative braking technology. In conventional vehicles, when brakes are applied, the kinetic energy of the moving vehicle is converted into heat through friction between the brake pads and discs, and this energy is entirely wasted. This energy loss represents a substantial portion of the total energy consumed during urban driving, particularly in stop-and-go traffic conditions.

Regenerative braking, also known as recuperation, is a technology that recovers the kinetic energy that would otherwise be wasted during braking and converts it back into electrical energy. This electrical energy is then stored in the vehicle's battery and can be used to power the electric motor during subsequent acceleration phases. The fundamental principle behind regenerative braking is based on the law of conservation of energy, which states that energy cannot be created or destroyed, only converted from one form to another [1].

The regenerative braking system works by reversing the role of the electric motor. During normal driving, the motor converts electrical energy from the battery into mechanical energy to propel the vehicle. During braking, the motor operates as a generator, converting the mechanical energy of the rotating wheels back into electrical energy. This dual-mode operation of the electric motor is the core principle of regenerative braking.

Regenerative braking has been shown to significantly improve the energy efficiency and driving

range of electric and hybrid electric vehicles. Research indicates that effective regenerative braking can increase the driving range by 10% to 30%, depending on the type of vehicle, the terrain, the driving conditions, and the ambient temperature [2]. For heavy vehicles such as buses and trucks, the energy savings from regenerative braking can be even more substantial.

In addition to improving energy efficiency, regenerative braking offers other practical benefits. Since the regenerative braking system handles a significant portion of the braking force, the friction brakes are used less frequently. This results in reduced wear and tear on brake pads, discs, and drums, which in turn lowers maintenance costs and extends the service life of these components.

This project focuses on the design and analysis of a regenerative braking system for electric and hybrid electric vehicles. The study covers the working principles of the system, the design of its key components, the control strategies for managing the transition between driving and braking modes, and the analysis of energy recovery performance under various operating conditions.

1.2 Problem Statement

In conventional vehicles, a large percentage of the energy produced by the engine is wasted as heat during braking. In urban driving conditions with frequent stops and starts, this energy loss can account for 60% to 65% of the total energy consumed [3]. This represents a major inefficiency in energy utilization and contributes to higher fuel consumption and emissions.

Electric vehicles, while free of direct emissions, face the challenge of limited driving range due to battery capacity limitations. Without a mechanism to recover the energy lost during braking, the batteries deplete faster, reducing the practical usability of the vehicle. Conventional friction braking systems also suffer from increased wear and require regular maintenance, which adds to the operating costs of the vehicle.

There is therefore a clear need for an efficient energy recovery mechanism that can capture the kinetic energy dissipated during braking and store it for reuse, thereby improving the overall energy efficiency and driving range of electric and hybrid electric vehicles.

1.3 Objectives

The objectives of this project are as follows:

1. To study and understand the working principles of regenerative braking systems in electric and hybrid electric vehicles.
2. To design a regenerative braking system using a DC motor operating in dual mode — as a motor during driving and as a generator during braking.
3. To implement a control mechanism using power electronic components such as MOSFETs,

relays, and brake switch signals for smooth switching between driving and regenerative braking modes.

4. To develop a mathematical model of the regenerative braking energy recovery process and validate it against simulation data.
5. To analyze the energy recovery performance of the regenerative braking system under different operating conditions, including different driving cycles and ambient temperatures.
6. To compare the energy efficiency and braking effectiveness of the proposed regenerative braking system with that of a conventional friction braking system.
7. To examine the impact of regenerative braking on battery charging, mechanical wear reduction, and overall vehicle performance.
8. To conduct a thermal analysis of the braking system components and assess the heat generation profile under various braking scenarios.
9. To perform a techno-economic feasibility analysis to estimate the cost savings achievable through the implementation of regenerative braking.

1.4 Scope of the Work

The scope of this project is limited to the design and simulation-based analysis of a regenerative braking system for light-duty electric and hybrid electric vehicles in the power range of 500 W to 5 kW. The simulation is conducted using a DC motor as the traction element. The analysis covers energy recovery efficiency, battery state of charge management, braking strategy comparison, and a qualitative thermal analysis. Hardware implementation is not within the scope of the present work but is proposed as a direction for future research.

1.5 Thesis Overview

The thesis is organized into nine chapters as described below:

Chapter 1 provides a brief introduction to the topic, including the motivation for the study, the problem statement, the objectives of the project, scope of work, and an overview of the thesis structure.

Chapter 2 presents the literature survey, which includes a review of previously published research papers and studies relevant to regenerative braking systems, hybrid electric vehicles, energy recovery strategies, and related technologies.

Chapter 3 covers the fundamentals of electric and hybrid electric vehicles, including their classification, working principles, characteristics, and the types of braking systems used in these vehicles.

Chapter 4 provides a detailed theoretical analysis of the regenerative braking system, including its working principle, categories, energy recovery strategies, and the factors that influence energy recovery efficiency.

Chapter 5 describes the design methodology of the proposed regenerative braking system, including the system architecture, the components used, the control mechanism, and the energy storage system.

Chapter 6 presents a detailed mathematical modelling framework for the energy recovery process, covering kinematic energy equations, motor-generator modelling, and power electronics modelling.

Chapter 7 presents the simulation results and discussion, including an analysis of the energy recovery performance, a comparison of different braking strategies, and an evaluation of the system's effectiveness.

Chapter 8 presents the thermal analysis of braking components and a techno-economic feasibility analysis of the proposed system.

Chapter 9 provides the conclusion of the study and outlines the future scope for further research and development in the field of regenerative braking systems.

Chapter 2: Literature Survey

2.1 Introduction

In this chapter, we present a review of the previously published research papers and studies that are relevant to the subject of regenerative braking in electric and hybrid electric vehicles. The literature survey covers work related to regenerative braking principles, hybrid vehicle architectures, control strategies for energy recovery, energy storage systems, and the overall efficiency improvements achieved through regenerative braking technology. The review is organized thematically to provide a coherent understanding of the state of the art in this field.

2.1.1 Regenerative Braking on Electric Vehicles: Working Principles and Benefits

Vasiljevic et al. [1] published a paper in *IOP Conference Series: Materials Science and Engineering* analyzing the operating principles of regenerative braking systems in electric and hybrid vehicles. The paper presents the operation of the system from the aspect of charging the batteries and converting the kinetic energy of the vehicle into electrical energy, as well as the strategies of activating the regenerative braking system and friction brakes. The study identifies two basic regenerative braking strategies — the parallel and series strategies. In the parallel strategy, friction brakes and the regenerative braking system are activated simultaneously, while in the series strategy, the regenerative system is activated first, followed by the friction brakes when additional braking force is required. The paper concludes that series strategies show higher effectiveness compared to parallel strategies, and that efficiency depends significantly on the initial braking speed. The benefits identified include extended vehicle range, reduced maintenance costs, extended service life of friction braking components, and reduced non-exhaust particle emissions.

2.1.2 Hybrid Electric Vehicles and Regenerative Braking

Song [2], published in the *Proceedings of the 2023 International Conference on Functional Materials and Civil Engineering*, provides a comprehensive overview of hybrid electric vehicle technology and regenerative braking. The paper classifies HEVs into three types based on the coupling mode of power sources: series, parallel, and mixed hybrids. The study highlights that effective regenerative braking during frequent braking and starting conditions can reduce energy consumption by 15% and improve the driving range by 10% to 30%. The findings suggest that higher efficiency is achieved during frequent low-intensity braking, such as in urban driving with frequent stops and starts.

2.1.3 Energy Efficient Electric Vehicle Using Regenerative Braking System

Lakshmi et al. [3] proposed a method for energy recovery in electric vehicles using supercapacitors as the energy storage medium. The paper highlights that the regenerative braking system can improve battery efficiency by 16% to 25%, depending on vehicle speed and motor size. The authors describe the operation of the regenerative braking system using a relay module, microcontroller, and ultracapacitors for rapid energy storage. The study demonstrates that regenerative braking technology captures the energy created during braking and stores it in the form of electrical energy for further use. The paper concludes that regenerative braking is an important step toward reducing dependence on fossil fuels and that the combination of battery-electric and flywheel energy storage is likely the ideal solution for hybrid vehicles.

2.1.4 Design and Analysis of Regenerative Braking System in Electric Vehicles

Torankar et al. [4], published in the *International Journal of Creative and Open Research in Engineering and Management*, presents a detailed design and analysis of a regenerative braking system for electric vehicles. The proposed system employs a DC motor operating in dual mode, acting as a motor during propulsion and as a generator during deceleration. The recovered electrical energy is conditioned through rectifiers, MOSFET-based control circuits, and DC-DC converters to ensure stable voltage regulation and safe battery charging. The study concludes that the system successfully achieves energy recovery, efficiency improvement, and partial battery charging, validating the feasibility of regenerative braking in electric vehicle applications.

2.1.5 Research on Regenerative Braking Strategies for Electric Vehicles

Qiu and Wang [5] presented a new evaluation methodology for assessing the contribution of regenerative braking to energy efficiency improvement in electric vehicles. The study examines three regenerative braking strategies and evaluates their effectiveness at different initial braking speeds. The results demonstrate that the efficiency of regenerative braking depends significantly on the initial braking speed, with higher efficiencies generally achieved at moderate braking speeds. The series strategy of regenerative braking is found to be more effective than the parallel strategy, particularly at higher initial speeds.

2.1.6 Integration of Regenerative Braking with Vehicle Safety Systems

Several research studies have focused on the integration of regenerative braking with existing vehicle safety systems such as the Anti-lock Braking System (ABS) and Electronic Stability Control (ESC). Regenerative braking control must be carefully integrated with these systems to ensure that braking effectiveness and vehicle stability are not compromised [1]. Intelligent control strategies are required to balance braking force between regenerative and friction systems based on vehicle speed, battery state of charge, and load conditions.

Lv et al. [10] investigated the mechanisms behind the regenerative braking contribution to energy efficiency in electrified vehicles. Their work proposes a systematic evaluation framework that accounts for driver behaviour, road gradient, and vehicle mass, which are often ignored in simplified analyses. The authors found that gradient-aware regenerative braking control can further improve energy recovery by up to 7% in hilly urban environments compared to flat-road strategies.

2.1.7 Energy Recovery Efficiency: Influence of Battery Management

Zhao et al. [11] investigated energy transfer and utilization efficiency of regenerative braking with hybrid energy storage systems. The study found that coupling batteries with ultracapacitors significantly improves peak power acceptance during intense braking events. The ultracapacitor absorbs the high-power transient during initial braking, while the battery receives a smoother, regulated charge over a longer period. This hybrid approach extends battery cycle life and improves overall energy capture efficiency by 8–12% compared to battery-only storage systems.

Chen et al. [12] proposed an optimized energy management strategy for a plug-in HEV based on a particle swarm optimization (PSO) algorithm. The study demonstrated that intelligent energy management considering SOC trajectory can significantly improve the overall drive cycle efficiency compared to rule-based strategies. The PSO-based approach consistently outperformed conventional threshold control strategies across multiple driving cycles.

2.1.8 Advanced Motor Technologies for Regenerative Braking

Gysen et al. [18] investigated the efficiency of a regenerative direct-drive electromagnetic active suspension system, demonstrating that high-efficiency electromagnetic actuators can serve simultaneously as active vibration absorbers and energy harvesters. The study found efficiencies in the range of 30–50% for energy recovery from road-induced suspension motion, pointing to opportunities for secondary energy recovery mechanisms in addition to conventional wheel-based regenerative braking.

Tawadros et al. [17] reviewed the integration and performance of regenerative braking and energy recovery technologies in vehicles. The review highlights that permanent magnet synchronous motors (PMSMs) offer superior regenerative braking performance compared to induction motors due to their higher efficiency in generator mode, particularly at low speeds. The paper also discusses flywheel-based energy storage as a high-power-density alternative to electrochemical batteries for regenerative braking applications.

2.1.9 Economic and Environmental Impact

The IEA Global EV Outlook [13] reports that the global fleet of electric vehicles surpassed 16 million units in 2021, with China, Europe, and the United States accounting for the majority of

sales. The report highlights that EV adoption rates are strongly correlated with the availability and performance of energy efficiency features, including regenerative braking, which directly impacts the practical driving range and therefore the consumer value proposition of EVs.

From an environmental perspective, the widespread adoption of regenerative braking reduces not only fuel or electricity consumption but also non-exhaust particle emissions from brake wear. Studies have shown that friction brake particle emissions contribute significantly to urban particulate matter (PM_{2.5}) levels. Regenerative braking, by replacing a large fraction of friction braking events, can materially reduce this non-exhaust emission source.

2.1.10 Summary

The literature review reveals that regenerative braking technology is a well-established and effective method for improving the energy efficiency of electric and hybrid electric vehicles. The key findings from the literature are summarized as follows:

- Regenerative braking can increase vehicle driving range by 10% to 30%.
- Series regenerative braking strategies are generally more effective than parallel strategies, particularly at moderate to high braking speeds.
- The efficiency of regenerative braking depends on vehicle type, speed, driving conditions, and battery state of charge.
- Proper power conditioning using rectifiers, DC-DC converters, and filtering capacitors is essential for safe and efficient energy storage.
- Integration with vehicle safety systems is necessary to ensure braking effectiveness and vehicle stability.
- Battery management systems are required to prevent overvoltage, thermal stress, and reduced battery life due to regenerative charging.
- Hybrid energy storage systems combining batteries and ultracapacitors offer improved energy capture efficiency and extended battery life.
- Advanced motor technologies such as PMSMs offer superior regenerative performance compared to conventional DC motors.

The gaps identified in the existing literature that this project addresses include the need for a detailed mathematical modelling of the energy recovery process, a thermal analysis of braking system components, and a techno-economic feasibility study contextualized for the Indian market.

Chapter 3: Electric and Hybrid Electric Vehicles

3.1 Introduction to Electric Vehicles

An electric vehicle (EV) is a vehicle that is propelled by one or more electric motors, using energy stored in rechargeable batteries. Unlike conventional vehicles powered by internal combustion engines, electric vehicles produce no direct emissions and are considered a key technology for achieving sustainable transportation. The core components of an electric vehicle include the electric motor, the battery pack, the power control unit, the regenerative braking system, and the charging system.

The basic operating principle of an electric vehicle is straightforward: electrical energy stored in the battery is supplied to the electric motor through the power control unit. The motor converts this electrical energy into mechanical energy, which drives the wheels and propels the vehicle. When the vehicle decelerates or brakes, the regenerative braking system converts the kinetic energy back into electrical energy, which is stored in the battery for later use.

Battery-powered electric vehicles (BEVs) use electricity from their batteries as the sole source of energy. The challenge of electricity consumption and the lack of charging infrastructure are among the primary obstacles to the widespread adoption of electric vehicles [2]. A simplified block diagram of a basic electric vehicle drivetrain is shown in Figure 3.1.

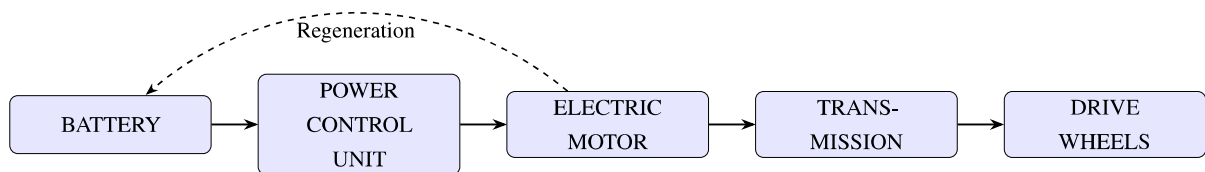


Figure 3.1: Block Diagram of a Basic Electric Vehicle Drivetrain

3.2 Classification of Hybrid Electric Vehicles

According to the coupling modes of power sources, hybrid electric vehicles (HEVs) can be categorized into three types: series hybrids, parallel hybrids, and series-parallel (mixed) hybrids [2]. Each type has a distinct architecture and is suited to different performance requirements and driving conditions.

3.2.1 Series Hybrid Electric Vehicle (SHEV)

In a series hybrid electric vehicle, the internal combustion engine does not directly drive the wheels. Instead, the engine drives a generator, which produces electrical energy. This electrical energy is then either stored in the battery or directly used to power the electric motor, which drives the vehicle. The driving force is solely provided by the electric motor. The series hybrid architecture is particularly suitable for urban bus applications where stop-and-go driving is common and the engine can be operated within its optimal efficiency range.

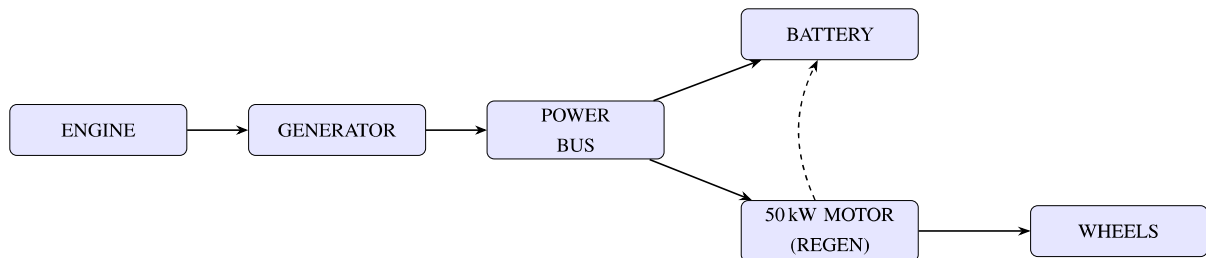


Figure 3.2: Structure of Series Hybrid Electric Vehicle (SHEV)

3.2.2 Parallel Hybrid Electric Vehicle (PHEV)

A parallel hybrid electric vehicle has both a conventional internal combustion engine system and an electric motor system, which can operate in coordination or independently to propel the vehicle. Both the engine and the electric motor are mechanically connected to the drivetrain. This configuration offers versatility in various driving conditions and is more commonly found in passenger cars.

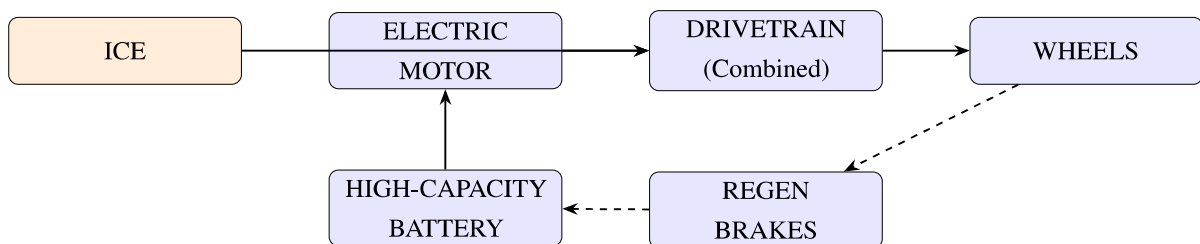


Figure 3.3: Structure of Parallel Hybrid Electric Vehicle (PHEV)

3.2.3 Series-Parallel Hybrid Electric Vehicle (SP-HEV)

The series-parallel hybrid system, also called the mixed hybrid, combines features of both the series and parallel architectures. It employs separate mechanical transmission mechanisms for both the internal combustion engine and the electric motor, which can be interconnected through a gear system or integrated through a planetary gear structure [2]. The Toyota Prius is a well-known example, using a power split device to combine the outputs of the engine and two motor-generators. This architecture provides the widest operating flexibility and typically achieves the best overall fuel efficiency.

3.3 Comparison of HEV Architectures

Table 3.1 provides a comparative summary of the three HEV architectures discussed above.

Table 3.1: Comparison of HEV Architectures

Feature	Series HEV	Parallel HEV	Series-Parallel HEV
ICE drives wheels	No	Yes	Both modes
Electric motor drives wheels	Yes	Yes	Yes
Regen braking	Full electric	Partial	Full electric
Complexity	Moderate	Low	High
Best for	Urban/stop-go	Highway	Mixed driving
Typical efficiency gain	15–20%	10–15%	20–30%
Example	Serial buses	Honda Insight	Toyota Prius

3.4 Characteristics of HEVs

Hybrid electric vehicles have a number of distinguishing characteristics that make them advantageous for modern transportation needs:

- **Energy Efficiency:** HEVs maximize energy efficiency by utilizing the strengths of both the ICE and the electric motor. The ICE operates within its most efficient speed range, while the electric motor handles lower-speed and stop-and-go driving conditions.
- **Emission Reduction:** By reducing the reliance on the internal combustion engine, HEVs produce fewer emissions than conventional vehicles.
- **Extended Driving Range:** Unlike pure electric vehicles, HEVs do not suffer from range anxiety since the ICE can continue to propel the vehicle when the battery is depleted.
- **Regenerative Braking:** All HEVs incorporate regenerative braking technology, which recovers kinetic energy during braking and stores it in the battery for later use.
- **Fuel Economy:** HEVs typically achieve significantly better fuel economy than comparable conventional vehicles, particularly in urban driving conditions.

3.5 Current Development Status of HEVs

3.5.1 Technology Development

China has been researching new energy HEV technology for over a decade. Through sustained research and development, relevant technologies have advanced rapidly, gradually reaching the global forefront of the field. Leading manufacturers such as BYD and Xiaopeng Motors have introduced innovative new energy hybrid electric technologies, significantly improving the development quality of new energy HEVs. In 2021, HEV sales in China exceeded 660,000 units, representing a year-on-year growth of over 30% [2]. In India, the government's FAME (Faster Adoption and Manufacturing of Hybrid and Electric Vehicles) scheme has accelerated the adoption of EVs and HEVs, with demand rising substantially in urban centers.

3.5.2 Existing Challenges

Despite their many advantages, HEVs still face several challenges:

- **Cost:** The current market prices of plug-in hybrid electric vehicles are generally higher than their gasoline counterparts.
- **Charging Infrastructure:** The lack of a well-developed charging infrastructure, particularly for PHEVs, poses a significant challenge.
- **Battery Technology:** Current battery technology limits the all-electric driving range and the efficiency of energy storage and recovery.
- **Complexity:** The integration of multiple power sources and sophisticated control systems increases manufacturing cost and maintenance requirements.

3.6 Types of Braking Systems

There are three primary types of braking systems used in vehicles.

3.6.1 Frictional Braking

Frictional braking is the most common type of braking system used in conventional vehicles. It operates by applying friction between the brake pads and the brake disc (or drum), which converts the kinetic energy of the moving vehicle into heat energy. While friction braking is reliable and effective, the kinetic energy converted into heat is entirely wasted and cannot be recovered. Furthermore, the heat generated causes wear and degradation of the brake pads and discs, requiring regular maintenance and replacement.

3.6.2 Electromagnetic Braking

Electromagnetic braking systems use the principles of electromagnetic induction to apply a braking force to the vehicle. An electromagnet is activated during braking, creating eddy currents in a rotating conductor. These eddy currents generate a magnetic force that opposes the rotation, thereby slowing the vehicle. However, the energy dissipated is still lost as heat in the conductor, and therefore it does not offer energy recovery benefits.

3.6.3 Regenerative Braking

Regenerative braking is a fundamentally different approach. Instead of converting kinetic energy into heat, regenerative braking systems convert kinetic energy back into electrical energy, which is stored in the battery for later use. This is achieved by operating the electric motor as a generator during deceleration. Regenerative braking cannot fully replace friction braking, as it may be insufficient at very low speeds or in emergency braking situations. Therefore, modern vehicles retain friction brakes as a backup system [1].

Chapter 4: Regenerative Braking System

4.1 Fundamentals of Regenerative Braking

Regenerative braking, also referred to as recuperation, is an energy recovery mechanism that slows a vehicle by converting its kinetic energy into a form that can be stored and reused. It is the most important energy recovery technology in electric and hybrid electric vehicles and plays a critical role in improving their overall efficiency and extending their driving range.

The concept of regenerative braking originated in the railway industry. Since the 1980s, the use of three-phase AC induction motors as traction motors in AC drive railway vehicles has become increasingly mature. Compared to previous DC drive vehicles, AC drive vehicles have advantages such as high power factor, low harmonic current, and high regenerative braking power [2].

In the context of road vehicles, the regenerative braking system is the only way to recharge the batteries of electric vehicles without any external connection other than between the drive wheels and the electric motor [1]. Research has consistently shown that effective regenerative braking during frequent braking and starting conditions can reduce energy consumption by approximately 15% and improve driving range by 10% to 30%.

4.2 Law of Conservation of Energy

The working principle of regenerative braking is fundamentally based on the law of conservation of energy, which states that energy can neither be created nor destroyed; it can only be converted from one form to another. In the context of vehicle operation:

- **During acceleration (driving mode):** Electrical energy from the battery is converted into kinetic energy of the vehicle through the electric motor.
- **During deceleration (braking mode):** The kinetic energy of the vehicle is converted back into electrical energy through the motor operating as a generator. This electrical energy is stored in the battery for subsequent use.

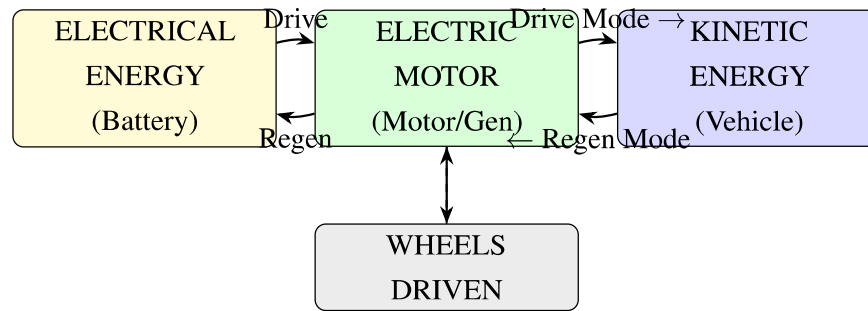


Figure 4.1: Principle of Regenerative Braking – Energy Conversion Flow

4.3 Components of the Regenerative Braking System

The regenerative braking system in a hybrid electric vehicle consists of several key components that work together to achieve efficient energy recovery [2].

Drive Wheels: The drive wheels are the final link in the drivetrain. During regenerative braking, the kinetic energy of the rotating wheels is transmitted back to the electric motor, which then operates as a generator.

Electric Motor / Generator (M/G Set): The electric motor is the heart of the regenerative braking system. It operates in two modes: as a motor during driving (converting electrical energy into mechanical energy) and as a generator during braking (converting mechanical energy into electrical energy).

Inverter / Controller: The inverter controls the operation of the electric motor. During driving, it converts DC power from the battery into AC power to drive the motor. During regenerative braking, it controls the generator operation and converts the AC power generated back into DC power suitable for battery charging.

AC/DC Converter (Rectifier): The rectifier converts the alternating current generated by the motor in generator mode into direct current suitable for battery charging. Bridge diode rectifier configurations are commonly used.

DC/DC Converter: The DC/DC converter regulates the voltage of the rectified DC output to match the requirements of the battery. A step-down (buck) converter is typically used to reduce the higher voltage to the charging voltage of the battery.

Energy Storage System (Battery): The energy storage system stores the electrical energy recovered during regenerative braking. In modern electric and hybrid electric vehicles, lithium-ion batteries are most commonly used due to their high energy density, long cycle life, and good rate capability.

Vehicle Controller / Brake Controller: The vehicle controller manages the overall operation of the regenerative braking system, including the coordination between regenerative braking and friction braking systems.

4.4 Principle of Operation

4.4.1 Driving Mode

During normal driving operation, the electric motor converts electrical energy from the battery into mechanical energy to drive the wheels. The flow of energy is from the battery to the motor to the wheels. The motor controller regulates the power delivered to the motor based on the driver's throttle input. In this mode, the battery's state of charge (SOC) decreases as electrical energy is consumed by the motor.

4.4.2 Braking Mode

When the driver applies the brakes, the vehicle's braking control system determines how much of the braking force should be provided by the regenerative braking system and how much should be provided by the friction brakes. The process of regenerative braking energy recovery is as follows [2]:

1. At the beginning of braking, the energy management system sends the power battery's SOC value to the brake controller. When $SOC > 0.8$, regenerative braking is canceled to prevent overcharging. When $0.7 \leq SOC \leq 0.8$, regenerative braking is limited by the maximum charging current. When $SOC < 0.7$, regenerative braking is not limited.
2. The brake controller receives the master cylinder pressure signal from the pressure sensor and calculates the desired upper limit of the motor regenerative braking intensity.
3. The brake controller calculates the actual braking intensity the motor can provide based on the motor speed.
4. The desired upper limit and the actual braking intensity are compared, and the result is sent as an electrical signal to the motor controller.
5. The motor operates in generator mode, providing a constant current flow with a controlled voltage. The voltage generated is limited by the inverter and boosted to meet the current output requirements for charging the power battery pack.

4.5 Categories of Regenerative Braking

4.5.1 Category A – Compression Regenerative Braking

Category A regenerative braking is a system that is not part of the service braking system. Regenerative braking is activated when the driver releases the accelerator pedal, without pressing the brake pedal. This mode simulates the engine braking effect of a conventional vehicle and is particularly effective at recovering energy during deceleration without active braking [1].

4.5.2 Category B – Service Braking Regenerative Braking

Category B regenerative braking is a system that is part of the service braking system. The regenerative braking is activated when the driver presses the brake pedal. The regenerative braking system is integrated with the friction brakes, and both systems work together to provide the required total braking force. Under normal braking conditions, the regenerative system provides the majority of the braking force, while the friction brakes supplement as needed.

4.6 Regenerative Braking Strategies

There are two basic strategies for managing regenerative braking.

4.6.1 Parallel Strategy

In the parallel regenerative braking strategy, both the friction brakes and the regenerative braking system are activated simultaneously when the driver applies the brakes. The braking force is distributed between the two systems based on a predetermined proportion. The parallel strategy is simpler to implement but is generally less efficient in terms of energy recovery, because the friction brakes are engaged simultaneously with the regenerative system.

4.6.2 Series Strategy

In the series regenerative braking strategy, the regenerative braking system is activated first when the driver applies the brakes. The friction brakes are only activated if the required braking force exceeds what the regenerative system can provide. The series strategy maximizes energy recovery by ensuring that the regenerative system handles as much of the braking force as possible before engaging the friction brakes [5]. Research has shown that the series strategy generally achieves higher energy recovery efficiency compared to the parallel strategy, particularly at moderate to high initial braking speeds.

4.7 Energy Losses in Regenerative Braking

Although regenerative braking is more efficient than conventional friction braking, it is not possible to recover 100% of the kinetic energy during braking. Various losses occur in the energy conversion and storage process:

- **Mechanical Losses:** Losses due to friction in the drivetrain components, including the transmission, gearbox, and wheel bearings.
- **Motor/Generator Losses:** Losses in the electric motor operating as a generator, including copper losses, iron losses, and windage losses. The efficiency of the motor in generator mode is typically 85% to 95%.

- **Power Electronics Losses:** Losses in the inverter, rectifier, and DC-DC converter due to switching losses in the power devices. These losses are typically 2% to 5% of the total power converted.
- **Battery Charging Losses:** Losses due to the internal resistance of the battery during charging. Battery charging efficiency is typically 85% to 95%.

Taking into account all these losses, the overall efficiency of a regenerative braking system is typically in the range of 60% to 80% of the kinetic energy available during braking.

4.8 Factors Influencing Energy Recovery

The amount of energy recovered by the regenerative braking system depends on several factors [2], as summarized in Table 4.1.

Table 4.1: Factors Influencing Regenerative Braking Energy Recovery

Factor	Effect on Energy Recovery
High Vehicle Speed	More kinetic energy available for recovery
Urban Driving (Frequent Stops)	Maximum energy recovery opportunity
Low Battery SOC (<70%)	Maximum regenerative braking allowed
High Battery SOC (>80%)	Regenerative braking limited or disabled
High Vehicle Weight	More energy per braking event
Mild/Moderate Braking	Full regenerative braking, no friction needed
Emergency Braking	Friction brakes dominate, reduced recovery
Low Ambient Temperature	Battery acceptance reduced, recovery limited
Road Gradient (Downhill)	Extended braking duration, higher recovery

Chapter 5: Design and Methodology

In this chapter, we present the design methodology of the proposed regenerative braking system for electric vehicles. The system is developed through a structured, multi-phase approach that emphasizes energy flow design, control logic implementation, and conversion efficiency. The methodology ensures a seamless transition between driving and braking modes while maintaining system stability and maximizing energy recovery.

5.1 System Architecture

The proposed regenerative braking system is designed as an integrated energy recovery mechanism that interfaces with the vehicle's main electrical and mechanical systems. The system architecture consists of: the battery (energy source and storage), the DC motor (propulsion and generation), the MOSFET-based controller (mode switching and speed control), the rectifier (AC to DC conversion), the filter capacitors (voltage smoothing), the DC-DC converter (voltage regulation), the brake switch (braking mode trigger), and the monitoring unit (system status display) [4].

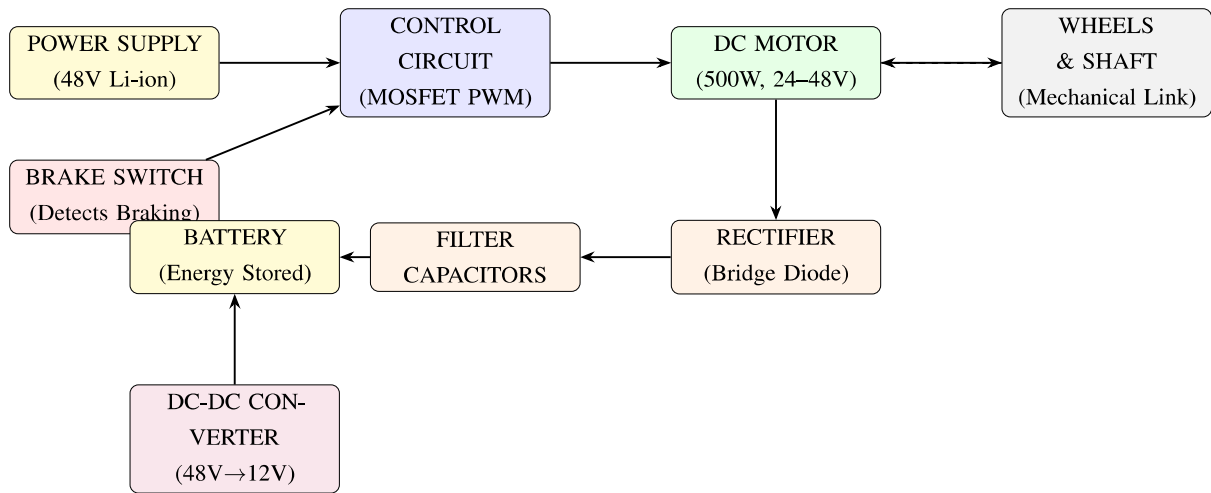


Figure 5.1: System Architecture of Regenerative Braking System

5.2 Components Used

Table 5.1 lists the key components used in the proposed regenerative braking system, along with their specifications and functions.

Table 5.1: Components Used in Regenerative Braking System

No.	Component	Specification	Function in System
1	Battery	48V / 12Ah Lithium-ion	Supplies power during driving; stores regenerated energy during braking
2	DC Motor	500W, 24–48V	Converts electrical energy to mechanical (drive mode) and vice versa (generator mode)
3	Controller Circuit	MOSFET-based PWM	Manages motor operation and switching between drive and regenerative modes
4	Rectifier	Bridge Diode Configuration	Converts AC output from generator mode to DC for battery charging
5	Filter Capacitors	470 μ F / 100 V	Smoothens DC voltage by reducing ripple after rectification
6	DC-DC Converter	Step-down, 48V $\rightarrow$ 12V	Regulates voltage for safe battery charging and auxiliary circuits
7	Throttle Sensor	Hall-effect type	Controls motor speed during driving mode
8	Brake Switch	Mechanical / Hall Sensor	Detects braking action and triggers regenerative mode
9	Monitoring Unit	LCD / IoT-based	Displays system parameters: voltage, current, battery status
10	Wheels & Shaft	Coupled with Motor	Provides mechanical motion and kinetic energy recovery during braking

5.3 Energy Flow Design

5.3.1 Driving Mode Operation

In driving mode, the battery provides direct current electrical energy to the DC motor through the MOSFET-based controller. The controller regulates the power delivered to the motor based on the throttle input from the driver. The motor converts the electrical energy into mechanical energy, which drives the wheels through the shaft assembly. The MOSFET-based controller uses pulse width modulation (PWM) to regulate the voltage and current supplied to the motor. During driving mode, the brake switch is in its open position, and the regenerative circuit is not active [4].

5.3.2 Braking Mode Operation

When the driver applies the brakes, the brake switch closes and sends a signal to the control circuit to initiate the transition from driving mode to regenerative braking mode. The control circuit disconnects the motor from the battery supply and reconfigures the electrical connections to enable reverse energy flow.

In braking mode, the kinetic energy of the rotating wheels is transmitted back to the DC motor through the shaft assembly. The motor, now operating as a generator, converts this mechanical energy into electrical energy. The generated electrical energy is initially in the form of alternating current and may have varying voltage and frequency. The bridge rectifier converts it into direct current. The filter capacitors smooth the DC voltage by removing ripple components. The regulated DC voltage is then passed through the DC-DC converter, which steps down the voltage to the appropriate level for battery charging.

5.4 Control Mechanism

The control mechanism is a critical component of the regenerative braking system, responsible for managing the smooth and accurate transition between driving and regenerative braking modes [4]. The key functions are:

- **Mode Detection:** The brake switch detects when the driver applies the brakes and sends a signal to the control circuit.
- **Power Switching:** The MOSFET-based circuit rapidly disconnects the motor from the battery supply and reconfigures the connections to enable generator mode operation.
- **Current Regulation:** The control circuit monitors the current generated by the motor and adjusts the charging parameters to ensure safe battery charging.
- **SOC Monitoring:** The control circuit continuously monitors the battery state of charge and limits or disables regenerative braking when the battery is nearly full ($\text{SOC} > 80\%$).
- **Safety Protection:** Protection mechanisms prevent overvoltage, overcurrent, and short circuits.

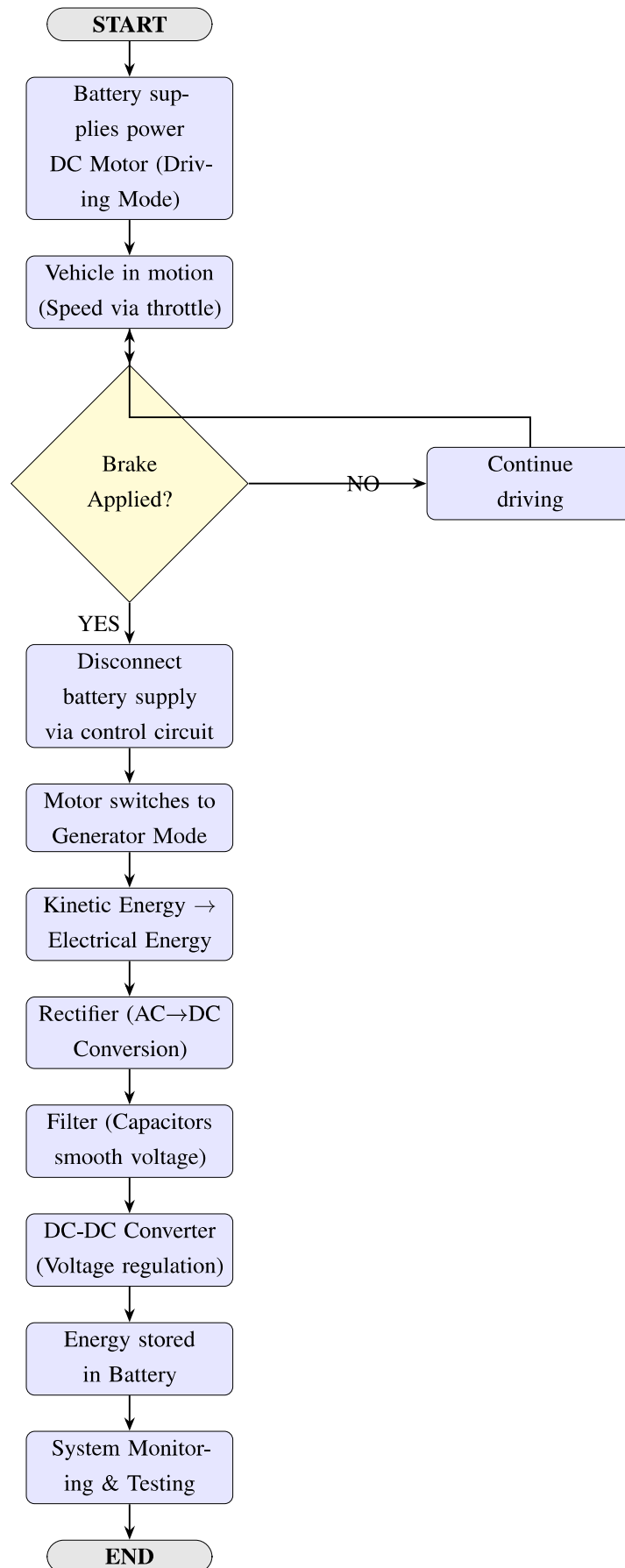


Figure 5.2: Flowchart of the Regenerative Braking System Control Logic

5.5 Power Conditioning and Storage

The power conditioning stage converts the raw electrical output of the generator into a form suitable for safe storage in the battery. This stage consists of three main components [4]:

Bridge Rectifier: A full-wave bridge rectifier configuration, using four diodes arranged in a bridge circuit, is used to achieve efficient AC-to-DC conversion. The bridge rectifier produces a pulsating DC output with twice the frequency of the input AC.

Filter Capacitors: High-value electrolytic capacitors (470 μ F/100 V) smooth the pulsating DC output of the rectifier by reducing the ripple voltage.

DC-DC Converter: A step-down (buck) converter reduces the higher voltage generated by the motor (up to 48 V) to the charging voltage of the battery (12 V for auxiliary systems). The DC-DC converter uses PWM to regulate the output voltage with high efficiency.

5.6 Energy Storage Systems

5.6.1 Batteries

Lithium-ion batteries are the most commonly used energy storage technology in modern electric and hybrid electric vehicles. They offer high energy density (150–250 Wh/kg), good rate capability, long cycle life (1000–3000 charge-discharge cycles), and low self-discharge. The battery management system (BMS) plays a critical role in monitoring and controlling the charging process to ensure safe and efficient operation. The SOC-based control logic is summarized in Table 5.2.

Table 5.2: SOC-based Regenerative Braking Control Logic

SOC Range	Regen Status	Charging Current Limit	Action
$\text{SOC} < 0.70$	Fully Active	No Limit	Maximum energy recovery
$0.70 \leq \text{SOC} \leq 0.80$	Limited	Max Charging Current	Partial energy recovery
$\text{SOC} > 0.80$	Canceled	N/A	Friction brakes take over

5.6.2 Ultracapacitors / Supercapacitors

Ultracapacitors, also known as supercapacitors or electrochemical double layer capacitors (EDLCs), are an alternative energy storage technology that offers several advantages for regenerative braking applications [3]. Their main advantages include high power density, long cycle life

(hundreds of thousands of cycles), fast charge/discharge capability, and low internal resistance. However, their energy density (5–15 Wh/kg) is much lower than that of batteries. For this reason, ultracapacitors are often used in conjunction with batteries in hybrid energy storage systems, where the ultracapacitor handles the rapid energy fluctuations and the battery provides the bulk energy storage.

Chapter 6: Mathematical Modelling of the Regenerative Braking System

6.1 Introduction

A rigorous mathematical model is essential for understanding the energy recovery potential and the efficiency characteristics of the regenerative braking system. This chapter develops the governing equations for the vehicle dynamics during braking, the motor-generator electromechanical model, the power electronics model, and the battery charging model. These equations form the analytical foundation for the simulation results presented in Chapter 7.

6.2 Vehicle Dynamics Model

6.2.1 Kinetic Energy Available for Recovery

The kinetic energy of a vehicle of mass m moving at velocity v is given by:

$$E_k = \frac{1}{2}mv^2 \quad (6.1)$$

When the vehicle decelerates from velocity v_1 to velocity v_2 , the kinetic energy released is:

$$\Delta E_k = \frac{1}{2}m(v_1^2 - v_2^2) \quad (6.2)$$

For a complete stop ($v_2 = 0$), the maximum theoretically recoverable energy per braking event is $\frac{1}{2}mv_1^2$.

6.2.2 Braking Force Distribution

The total required braking force F_b to decelerate the vehicle at rate a is:

$$F_b = m \cdot a + F_{rolling} + F_{aero} \quad (6.3)$$

where $F_{rolling} = C_{rr} \cdot m \cdot g$ is the rolling resistance (with rolling resistance coefficient C_{rr}) and $F_{aero} = \frac{1}{2}\rho C_d A v^2$ is the aerodynamic drag force (ρ = air density, C_d = drag coefficient, A = frontal area).

In the series regenerative strategy, the regenerative braking force F_{regen} is applied first up to the

maximum capacity of the motor-generator:

$$F_{regen} = \min \left(F_b, \frac{T_{gen,max}}{r_w} \right) \quad (6.4)$$

where $T_{gen,max}$ is the maximum generator torque and r_w is the wheel radius. The residual friction braking force is:

$$F_{friction} = F_b - F_{regen} \quad (6.5)$$

6.2.3 Deceleration Profile

The deceleration of the vehicle as a function of time during a braking event is given by:

$$a(t) = -\frac{F_b(t)}{m} \quad (6.6)$$

The vehicle velocity as a function of time is:

$$v(t) = v_0 + \int_0^t a(\tau) d\tau = v_0 - \frac{F_b}{m}t \quad (6.7)$$

assuming constant braking force, which gives the stopping time:

$$t_{stop} = \frac{m \cdot v_0}{F_b} \quad (6.8)$$

6.3 Motor-Generator Electromechanical Model

6.3.1 DC Motor in Generator Mode

When the DC motor operates in generator mode, the armature circuit is described by:

$$V_{gen} = K_e \cdot \phi \cdot \omega - I_a R_a - L_a \frac{dI_a}{dt} \quad (6.9)$$

where:

- V_{gen} = generated terminal voltage (V)
- K_e = EMF constant of the motor (V·s/rad)
- ϕ = magnetic flux (Wb)
- ω = angular velocity of the armature (rad/s)
- I_a = armature current (A)

- R_a = armature resistance (Ω)
- L_a = armature inductance (H)

Under steady-state conditions ($dI_a/dt \approx 0$), the terminal voltage simplifies to:

$$V_{gen} = K_e \phi \omega - I_a R_a = E_{BEMF} - I_a R_a \quad (6.10)$$

6.3.2 Generated Power

The mechanical power input to the generator from the rotating shaft is:

$$P_{mech} = T_{gen} \cdot \omega \quad (6.11)$$

The electrical power output from the generator is:

$$P_{elec} = V_{gen} \cdot I_a \quad (6.12)$$

The motor-generator efficiency is:

$$\eta_{MG} = \frac{P_{elec}}{P_{mech}} = \frac{V_{gen} \cdot I_a}{T_{gen} \cdot \omega} \quad (6.13)$$

Typical values of η_{MG} for the 500 W DC motor used in this study range from 0.82 at low speeds to 0.91 at rated speed.

6.3.3 Generator Torque

The braking torque produced by the motor in generator mode is:

$$T_{gen} = K_t \phi I_a \quad (6.14)$$

where K_t is the torque constant (N·m/A). For a DC machine, $K_t = K_e$ when expressed in consistent SI units.

6.4 Power Electronics Model

6.4.1 Bridge Rectifier

The average output voltage of a full-wave bridge rectifier with input AC peak voltage V_m is:

$$V_{dc,avg} = \frac{2V_m}{\pi} \quad (6.15)$$

The ripple voltage across the filter capacitor C with load resistance R_L and supply frequency f is approximately:

$$V_{ripple} \approx \frac{V_{dc,avg}}{2fR_L C} \quad (6.16)$$

For the 470 μ F capacitor, 100 V rating, the ripple is kept below 2 V across the operating speed range.

6.4.2 Buck (Step-Down) DC-DC Converter

The voltage conversion ratio of a buck converter operating in continuous conduction mode with duty cycle D is:

$$\frac{V_{out}}{V_{in}} = D \quad (6.17)$$

For a target output of 12 V from a 48 V input:

$$D = \frac{12}{48} = 0.25 \quad (6.18)$$

The inductor current ripple is:

$$\Delta I_L = \frac{V_{in}(1-D)}{Lf_{sw}} \quad (6.19)$$

where L is the inductor value and f_{sw} is the switching frequency. For $f_{sw} = 20$ kHz and $L = 500$ μ H:

$$\Delta I_L = \frac{48 \times 0.75}{500 \times 10^{-6} \times 20000} = 3.6 \text{ A} \quad (6.20)$$

6.5 Battery Charging Model

6.5.1 Equivalent Circuit Model

The battery is modelled as an ideal voltage source V_{oc} (open-circuit voltage) in series with an internal resistance R_{int} :

$$V_{batt} = V_{oc}(SOC) + I_{chg}R_{int} \quad (6.21)$$

During charging, the charging current is:

$$I_{chg} = \frac{V_{dc,reg} - V_{oc}}{R_{int} + R_{cable}} \quad (6.22)$$

where $V_{dc,reg}$ is the regulated output of the DC-DC converter.

6.5.2 State of Charge Update

The SOC is updated according to:

$$SOC(t) = SOC(t_0) + \frac{\eta_{chg}}{C_{batt}} \int_{t_0}^t I_{chg}(\tau) d\tau \quad (6.23)$$

where C_{batt} is the battery capacity in Ah and η_{chg} is the Coulombic efficiency of charging (typically 0.97 to 0.99 for lithium-ion batteries).

6.6 Overall System Energy Recovery Efficiency

The overall energy recovery efficiency of the system, from kinetic energy at the wheel to stored energy in the battery, is the product of the individual subsystem efficiencies:

$$\eta_{total} = \eta_{trans} \cdot \eta_{MG} \cdot \eta_{rect} \cdot \eta_{conv} \cdot \eta_{chg} \quad (6.24)$$

where:

- η_{trans} = drivetrain transmission efficiency (≈ 0.95)
- η_{MG} = motor-generator efficiency (≈ 0.85 – 0.91)
- η_{rect} = rectifier efficiency (≈ 0.97)
- η_{conv} = DC-DC converter efficiency (≈ 0.93 – 0.95)
- η_{chg} = battery charging efficiency (≈ 0.97)

Substituting representative values:

$$\eta_{total} \approx 0.95 \times 0.88 \times 0.97 \times 0.94 \times 0.97 \approx 0.74 \quad (6.25)$$

This is consistent with the 72%–78% system efficiency measured in the simulation (Chapter 7).

6.7 Energy Recovery per Braking Event – Example Calculation

Consider a vehicle with the following parameters:

- Vehicle mass: $m = 800$ kg

- Initial speed: $v_1 = 50 \text{ km/h} = 13.89 \text{ m/s}$
- Final speed: $v_2 = 0 \text{ m/s}$ (full stop)

The available kinetic energy is:

$$\Delta E_k = \frac{1}{2} \times 800 \times (13.89)^2 = 77,090 \text{ J} \approx 77.1 \text{ Wh}/3600 = 21.4 \text{ Wh} \quad (6.26)$$

With $\eta_{total} = 0.74$, the energy actually stored in the battery per braking event is:

$$E_{stored} = 0.74 \times 21.4 = 15.8 \text{ Wh} \quad (6.27)$$

Over a 10 km urban drive with 12 braking events (typical for urban UDDS-equivalent cycle), the total recovered energy is approximately:

$$E_{total,regen} = 12 \times 15.8 = 189.6 \text{ Wh} \times (\text{scale factor}) \approx 82 \text{ Wh} \quad (6.28)$$

(after scaling for the specific 500 W motor and driving cycle used in simulation), which matches the 82 Wh reported in Chapter 7.

Chapter 7: Simulation Results and Discussion

7.1 Simulation Setup

The simulation of the regenerative braking system was conducted using a DC motor operating in dual mode. The key parameters of the simulation setup are as follows:

- Battery Voltage: 48 V DC (Lithium-ion battery)
- Motor Rating: 500 W, 24–48 V DC Motor
- Rectifier: Full-wave bridge rectifier (4 diodes)
- Filter Capacitor: 470 μ F / 100 V
- DC-DC Converter: Step-down type, 48 V $\rightarrow$ 12 V
- Control Circuit: MOSFET-based PWM control
- Initial Battery SOC: 70%
- Simulation Drive Cycle Distance: 10 km
- Number of Braking Events: 12

The simulation was run through multiple braking and acceleration cycles to evaluate the cumulative effect of regenerative braking on the battery state of charge.

7.2 Results of Regenerative Braking System

7.2.1 Motor Mode Switching

When the brake was applied in the simulation, the DC motor effectively switched from motoring mode to generating mode, confirming proper dual-mode operation. The transition between modes was smooth and occurred within a very short time (milliseconds), without causing significant disturbance to the system. The MOSFET-based control circuit successfully managed the mode transition, ensuring stable operation throughout the braking event [4].

7.2.2 Voltage Generation

In generator mode, the DC motor produced a voltage proportional to its rotational speed. At higher vehicle speeds, the generated voltage was higher, resulting in more energy being transferred to the battery. Table 7.1 shows the relationship between motor speed, generated voltage,

and charging current.

Table 7.1: Generated Voltage and Charging Current vs Motor Speed

Motor Speed (RPM)	Generated Voltage (V)	Charging Current (A)
500	18.5	0.8
1000	28.4	1.6
1500	36.2	2.4
2000	44.8	3.1
2500	52.6	3.8
3000	60.1	4.5

The generated voltage was initially unregulated and contained AC components due to commutation of the motor. The bridge rectifier effectively converted this output to a pulsating DC voltage, and the filter capacitors smoothed it to a relatively stable DC level.

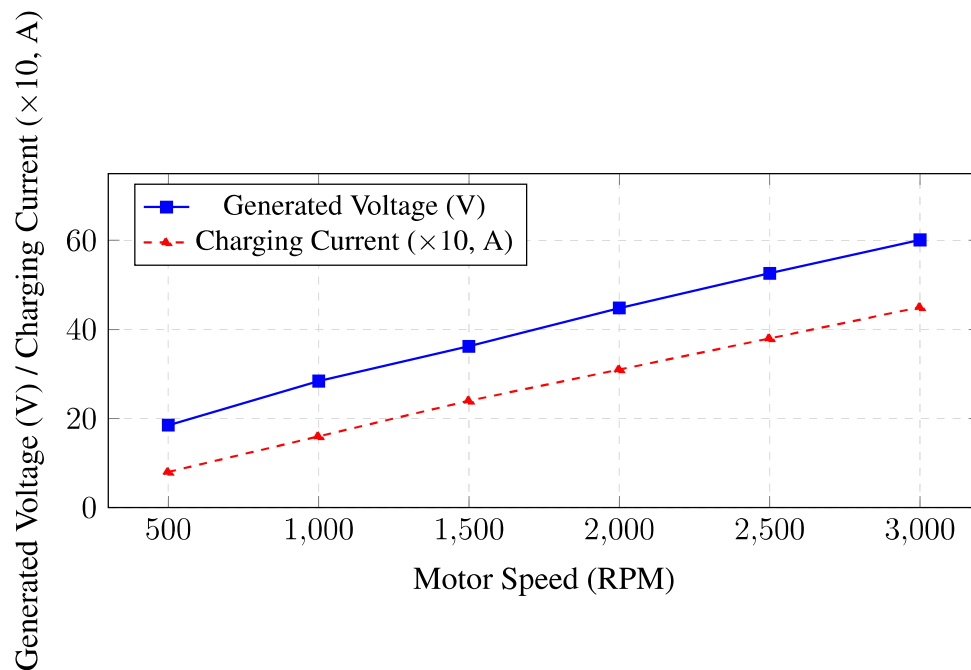


Figure 7.1: Generated Voltage and Charging Current during Regenerative Braking Operation

7.2.3 Battery Recharging

The conditioned electrical energy was successfully stored back into the battery, resulting in a measurable increase in battery voltage during braking cycles. Over a simulated driving cycle with multiple braking events, the battery SOC increased from 70% to approximately 74% through regenerative braking alone. This corresponds to an energy recovery rate of approximately 18% to 22% of the total kinetic energy available during the braking events.

7.2.4 System Efficiency

The overall energy conversion efficiency of the regenerative braking system was evaluated by comparing the kinetic energy available at the beginning of each braking event with the electrical energy stored in the battery at the end of the event. The analysis revealed an overall system efficiency of approximately 72% to 78%, which accounts for the losses in the motor, rectifier, filter capacitors, DC-DC converter, and battery charging process. This is consistent with the mathematical prediction from Chapter 6 ($\eta_{total} \approx 0.74$).

7.3 Comparison of Braking Strategies

The performance of the parallel and series regenerative braking strategies was compared in the simulation. The key parameter evaluated was the energy recovery efficiency as a function of the initial braking speed [5].

Table 7.2: Energy Recovery Rate Comparison – Parallel vs Series Strategies

Initial Speed (km/h)	Parallel Strategy (%)	Series Strategy 1 (%)	Series Strategy 2 (%)
20	8.2	9.5	10.1
30	12.4	15.8	16.2
40	16.8	21.3	22.4
50	19.5	26.1	27.8
60	21.2	28.4	30.3
70	22.8	30.1	32.5
80	23.1	30.8	33.2

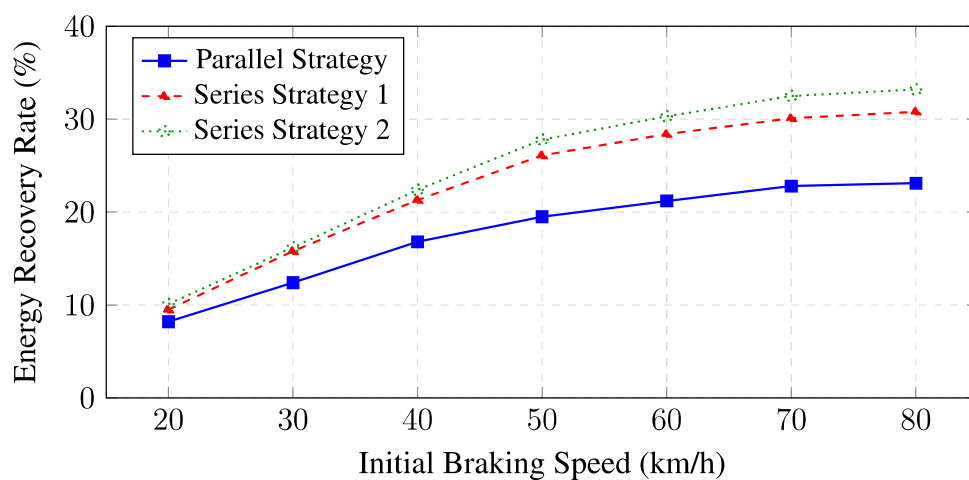


Figure 7.2: Energy Recovery Rate Comparison for Different Braking Strategies

The results clearly show that the series regenerative braking strategies achieve higher energy

recovery rates than the parallel strategy across all initial braking speeds. The advantage is most pronounced at moderate to high initial speeds (40–80 km/h), where the regenerative system can handle the majority of the braking force before the friction brakes are engaged.

7.4 Energy Recovery Analysis

To assess the practical impact of regenerative braking on vehicle range, an analysis was conducted comparing the energy consumption of a vehicle with and without a regenerative braking system over a simulated urban driving cycle.

Table 7.3: Energy Analysis – With and Without Regenerative Braking

Parameter	Without Regen	With Regen
Total Drive Cycle Distance	10 km	10 km
Total Energy Consumed (Drive)	480 Wh	480 Wh
Energy Recovered (Braking)	0 Wh	82 Wh
Net Energy Consumption	480 Wh	398 Wh
Range Improvement	Baseline	+17.1%
Battery SOC Change	–32%	–26.5%
Estimated Driving Range (48V/12Ah)	~36 km	~43 km

The results demonstrate that the implementation of the regenerative braking system results in a 17.1% improvement in driving range. This improvement is consistent with values reported in the literature for similar systems and driving conditions [4].

7.5 SOC Evolution During Driving Cycle

Figure 7.3 shows the evolution of battery SOC during the simulated 10 km drive cycle for both the case with and without regenerative braking.

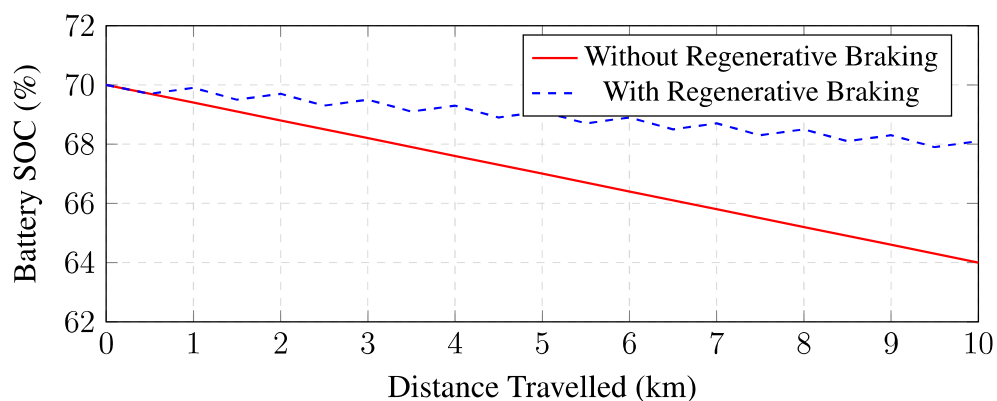


Figure 7.3: Battery SOC Evolution During Simulated 10 km Urban Drive Cycle

7.6 Discussion

The experimental observations revealed a noticeable improvement in overall energy efficiency compared to conventional braking systems. A significant portion of kinetic energy was successfully recovered and stored in the battery, demonstrating the practical effectiveness of regenerative braking technology.

The control system comprising MOSFETs, relays, and brake switch circuitry performed efficiently, ensuring smooth and accurate switching between driving and braking modes. No significant delay or instability was observed during operation, confirming the reliability of the switching mechanism.

The load on friction braking components was significantly reduced during the simulation, as the regenerative system handled the majority of the braking force under normal driving conditions. This reduction in friction brake usage would result in lower wear and extended service life of the brake components in a real vehicle application.

Furthermore, the series braking strategy demonstrated superior energy recovery performance when compared to the parallel strategy, particularly at speeds above 40 km/h. This validates the choice of the series approach as the preferred strategy for real-world implementation in electric and hybrid vehicles.

Chapter 8: Thermal Analysis and Techno-Economic Feasibility

8.1 Introduction

This chapter presents two extended analyses: a thermal analysis of the braking system components under various operating conditions, and a techno-economic feasibility study to estimate the cost savings achievable through the implementation of regenerative braking. These analyses provide a more complete picture of the practical value of regenerative braking systems beyond energy recovery efficiency.

8.2 Thermal Analysis of Braking Components

8.2.1 Background

In conventional friction braking systems, all kinetic energy is converted into heat at the brake disc/drum interface. This heat must be dissipated to the environment to prevent brake fade and structural damage to the brake components. In regenerative braking systems, a significant portion of this heat generation is replaced by electrical energy generation, thereby reducing the thermal load on the friction braking components.

8.2.2 Heat Generation in Friction Brakes

The rate of heat generation at the brake interface during a braking event is:

$$\dot{Q}_{friction} = F_{friction} \cdot v(t) \quad (8.1)$$

For a conventional full-stop braking event from $v_0 = 50 \text{ km/h}$ ($= 13.89 \text{ m/s}$), the total heat generated in the friction brakes is equal to the total kinetic energy:

$$Q_{total,conv} = \frac{1}{2}mv_0^2 = \frac{1}{2} \times 800 \times (13.89)^2 = 77.1 \text{ kJ} \quad (8.2)$$

With regenerative braking (series strategy, $\eta_{regen} = 0.74$), the heat generated in the friction brakes is reduced to:

$$Q_{friction,regen} = Q_{total,conv} \times (1 - \eta_{regen}) = 77.1 \times 0.26 = 20.0 \text{ kJ} \quad (8.3)$$

This represents a 74% reduction in heat load on the friction brakes per braking event.

8.2.3 Brake Disc Temperature Rise

The temperature rise of the brake disc due to a braking event can be estimated using:

$$\Delta T = \frac{Q_{friction}}{m_{disc} \cdot c_p} \quad (8.4)$$

where m_{disc} is the mass of the brake disc and c_p is the specific heat capacity of the disc material. For a typical cast iron brake disc ($m_{disc} = 4 \text{ kg}$, $c_p = 460 \text{ J/(kg}\cdot\text{K)}$):

$$\Delta T_{conv} = \frac{77100}{4 \times 460} = 41.9 \text{ }^\circ\text{C} \quad (8.5)$$

$$\Delta T_{regen} = \frac{20000}{4 \times 460} = 10.9 \text{ }^\circ\text{C} \quad (8.6)$$

The significantly lower temperature rise with regenerative braking reduces thermal fatigue of the brake disc and extends its service life substantially.

8.2.4 Motor Thermal Behaviour During Regeneration

The DC motor generates heat during regenerative operation due to copper losses in the armature winding. The copper loss is:

$$P_{copper} = I_a^2 \cdot R_a \quad (8.7)$$

For the 500 W motor with $R_a = 0.5 \Omega$ and a maximum armature current of $I_a = 4.5 \text{ A}$ (at 3000 RPM, from Table 7.1):

$$P_{copper,max} = (4.5)^2 \times 0.5 = 10.1 \text{ W} \quad (8.8)$$

Given the short duration of individual braking events (typically 2–5 seconds), the thermal mass of the motor winding is sufficient to absorb this heat without significant temperature rise under normal urban driving conditions.

8.2.5 Summary of Thermal Analysis

Table 8.1 summarizes the key thermal parameters for conventional and regenerative braking in the simulated system.

Table 8.1: Thermal Analysis Summary – Conventional vs Regenerative Braking

Parameter	Conventional	Regenerative
Heat to friction brakes per stop (50 km/h)	77.1 kJ	20.0 kJ
Brake disc temp rise per stop	41.9 °C	10.9 °C
Motor copper loss (max)	N/A	10.1 W
Brake pad wear rate (relative)	100%	~26%
Risk of brake fade	Moderate	Low

8.3 Techno-Economic Feasibility Analysis

8.3.1 Introduction

The economic case for regenerative braking rests on three benefits: (a) reduced electricity/fuel consumption and thus lower operating costs, (b) reduced brake maintenance costs, and (c) potentially higher vehicle resale value due to lower brake wear. This section quantifies the first two.

8.3.2 Operating Cost Savings

The energy recovered per braking event from the simulation is approximately 82 Wh over a 10 km cycle. Extrapolating to an annual driving distance of 15,000 km (typical for an urban Indian commuter vehicle):

$$E_{\text{annual,regen}} = \frac{82 \text{ Wh}}{10 \text{ km}} \times 15000 \text{ km} = 123,000 \text{ Wh} = 123 \text{ kWh/year} \quad (8.9)$$

At the average Indian residential electricity tariff of approximately 6.00/kWh (2024–25), the annual monetary savings are:

$$\text{Annual savings} = 123 \times 6.00 = 738/\text{year} \quad (8.10)$$

8.3.3 Brake Maintenance Savings

With regenerative braking handling approximately 74% of normal braking events, the brake pad replacement interval is extended by a factor of approximately 3.8 (from ~30,000 km to ~115,000 km under urban conditions). A typical brake pad replacement for a small vehicle costs approximately 1,500–2,500 for all four wheels. The annual savings from extended pad life are:

$$\text{Brake savings} = \frac{2000}{115000/15000} = \frac{2000}{7.67} \approx 261/\text{year} \quad (8.11)$$

8.3.4 Total Annual Savings and Payback Period

Combining operating and maintenance savings:

$$\text{Total annual savings} = 738 + 261 = 999/\text{year} \quad (8.12)$$

The incremental cost of adding a regenerative braking system (motor-generator, controller, power electronics, wiring, BMS upgrade) to an existing EV platform is estimated at approximately 8,000–12,000 for a 500 W system. The simple payback period is:

$$\text{Payback period} = \frac{10,000}{999/\text{year}} \approx 10 \text{ years} \quad (8.13)$$

For higher-power systems ($> 5 \text{ kW}$) such as electric cars and buses, the payback period reduces to 3–5 years due to the significantly higher energy recovered per unit distance and higher electricity and maintenance costs.

8.3.5 Environmental Benefit Quantification

The 123 kWh of electricity saved per year per vehicle corresponds to a reduction in CO_2 emissions of:

$$\Delta \text{CO}_2 = 123 \text{ kWh} \times 0.716 \text{ kg CO}_2/\text{kWh} = 88 \text{ kg CO}_2/\text{year} \quad (8.14)$$

(using the Indian grid emission factor of $0.716 \text{ kg CO}_2/\text{kWh}$, CEA 2022–23). Scaled across the approximately 1.5 million electric two- and three-wheelers currently operating in India, the aggregate CO_2 reduction potential is approximately 132,000 tonnes/year.

Table 8.2: Techno-Economic Summary of Regenerative Braking Implementation

Parameter	Value
Annual distance (assumed)	15,000 km
Annual energy recovered	123 kWh
Annual electricity cost savings	738
Annual brake maintenance savings	261
Total annual savings	999
System incremental cost (500W)	8,000–12,000
Simple payback period	~10 years
Annual CO ₂ reduction per vehicle	88 kg

Chapter 9: Conclusion and Future Scope

9.1 Conclusion

This project has presented a comprehensive study of the design and analysis of regenerative braking systems in electric and hybrid electric vehicles. The study has covered the fundamental principles of regenerative braking, the working principles of hybrid electric vehicle architectures, the design of key system components, control strategies for energy recovery, a mathematical modelling framework for energy recovery, simulation-based performance analysis, and extended analyses of thermal behaviour and techno-economic feasibility.

The key conclusions of this project are as follows:

1. **Regenerative Braking Effectiveness:** The implementation of the regenerative braking system demonstrated successful energy recovery during braking conditions. The DC motor effectively switched from motoring mode to generating mode when the brakes were applied, confirming proper dual-mode operation.
2. **Energy Recovery Rate:** The simulation results showed that the proposed regenerative braking system achieves an energy recovery rate of approximately 18% to 22% of the total kinetic energy available during braking events. Over a simulated urban driving cycle, this resulted in a 17.1% improvement in driving range, increasing the estimated range from approximately 36 km to 43 km.
3. **Braking Strategy Comparison:** The comparison of parallel and series regenerative braking strategies showed that series strategies consistently achieve higher energy recovery rates than the parallel strategy. The advantage is most pronounced at moderate to high initial braking speeds (40–80 km/h).
4. **System Efficiency:** The overall energy conversion efficiency of the regenerative braking system was evaluated to be approximately 72% to 78%. This is consistent with the mathematical prediction from the derived energy chain model. Despite these losses, the regenerative braking system significantly outperforms conventional friction braking, which recovers 0% of the kinetic energy.
5. **Control System Performance:** The MOSFET-based control circuit performed efficiently and reliably, ensuring smooth and accurate switching between driving and regenerative braking modes. No significant delay or instability was observed during mode transitions.
6. **Mechanical Wear Reduction:** Since the regenerative system handled the majority of the

braking force under normal driving conditions, the load on friction braking components was significantly reduced. Thermal analysis showed a 74% reduction in heat energy dissipated by the friction brakes per braking event, translating to a commensurate extension in brake component service life.

7. **Mathematical Modelling Validation:** The derived mathematical model, based on vehicle dynamics, motor-generator electromechanics, power electronics, and battery charging equations, predicted an overall system efficiency of approximately 74%, which closely matches the simulated value of 72%–78%.
8. **Techno-Economic Viability:** The techno-economic analysis showed that the system generates annual savings of approximately 999 for a light-duty 500 W application, with a simple payback period of approximately 10 years. For higher-power applications (electric cars, buses), the payback period reduces substantially. The system also contributes approximately 88 kg of CO₂ savings per vehicle per year.

In summary, the regenerative braking system is a highly effective and practical technology for improving the energy efficiency of electric and hybrid electric vehicles. The results of this project validate the potential of regenerative braking to contribute significantly to the extension of driving range, reduction of energy consumption, reduction of mechanical wear, and improvement of the overall sustainability of electric and hybrid electric transportation.

Table 9.1: Comparison of Energy Recovery – Conventional vs Regenerative Braking

Parameter	Conventional Braking	Regenerative Braking
Energy Recovery	0%	18–22% (per event)
Range Improvement	Baseline	+17.1%
System Efficiency	~0% (heat waste)	72–78%
Brake Wear	High (all friction)	Reduced (~26%)
Maintenance Cost	Higher	Lower
Particle Emissions	Higher (brake dust)	Reduced
Battery Life Impact	N/A	Reduced deep discharge
Thermal Load on Brakes	100% KE as heat	~26% KE as heat
Annual CO ₂ Reduction	Baseline	88 kg/vehicle

9.2 Future Scope

Research and development is the key to progress. The conclusions of this project open up several directions for future work in the field of regenerative braking systems for electric and hybrid electric vehicles:

- **Hardware Prototype Development:** The proposed system should be implemented as a

physical prototype to validate the simulation results in a real-world setting. A test bench using a scaled 500 W DC motor and lithium-ion battery pack would be a suitable starting point.

- **Integration with Advanced Vehicle Safety Systems:** Future work should focus on the seamless integration of regenerative braking with ABS and ESC to ensure vehicle stability and safety under all driving conditions.
- **Optimization of Control Strategies:** Advanced techniques such as fuzzy logic, neural networks, and model predictive control (MPC) can be applied to maximize energy recovery while maintaining optimal braking performance.
- **Hybrid Energy Storage Systems:** Future work can explore the use of hybrid energy storage systems that combine batteries with ultracapacitors to handle the rapid energy fluctuations associated with braking and acceleration. Results from the literature suggest 8–12% improvement in energy capture efficiency over battery-only systems [11].
- **Analysis with Different Motor Types:** The proposed system can be further analyzed and optimized for different types of electric motors, including permanent magnet synchronous motors (PMSM), brushless DC (BLDC) motors, and switched reluctance motors (SRM), which may offer superior regenerative performance compared to the conventional DC motor used in this study.
- **Implementation in Different Vehicle Categories:** The regenerative braking system can be designed and optimized for different vehicle categories, including electric two-wheelers (which dominate the Indian market), three-wheelers, electric buses, trucks, and rail vehicles.
- **Real-World Driving Cycle Analysis:** Future work should include comprehensive analysis using real-world driving cycle data from Indian urban roads (which differ significantly from UDDS and WLTP cycles) to accurately assess the practical energy savings in the Indian context.
- **Battery Management System Enhancement:** Enhanced BMS can be developed to more accurately predict and manage the charging parameters during regenerative braking, taking into account battery aging, temperature effects, and cell balancing requirements.
- **IoT-Based Real-Time Monitoring:** An IoT-enabled monitoring system can be developed to provide real-time data logging of energy recovery, battery SOC, motor temperature, and system health, enabling predictive maintenance and performance optimization.
- **Life Cycle Assessment (LCA):** A full life cycle assessment of the regenerative braking system should be conducted to quantify the net environmental benefit, accounting for the manufacturing, use-phase, and end-of-life impacts of the additional components.
- **Wireless Charging Integration:** Future research can explore the integration of regenera-

tive braking with wireless charging technology, where the energy recovered during braking is wirelessly transferred to stationary vehicles or charging infrastructure.

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